Improving the Safety of Medication Recommendation via Graph Augmented Patient Similarity Network

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Abstract

Recommending optimal medication combinations for patients is a crucial application of artificial intelligence in healthcare. Recent works typically use patients' electronic health record combined with their current health conditions. However, these efforts have the following issues: 1) they often reference historical visits unrelated to the current situation, and 2) there is a latent risk of side effects from historical prescriptions. Such issues raise concerns about the safety of medication recommendation. To address this, we propose **GPSRec**, a novel **G**raph augmented **P**atient **S**imilarity network for medication **Rec**ommendation. By leveraging dual similarity measures to selectively integrate historical visits, GPSRec effectively filters out irrelevant information, improving the accuracy of recommendation. We further present a training strategy, which combines a pre-training method and a dual threshold loss adjustment, reduces the risk of adverse drug-drug interactions, enhancing the safety of recommendation. Extensive experiment results on two real datasets demonstrate that GPSRec significantly outperforms state-of-the-art methods. Notably, it achieves 30.11% and 24.92% improvements in safety, respectively, with higher accuracy.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Applied computing** → **Health informatics**.

Keywords

Medication recommendation; Recommender systems; Electronic health record

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1 Introduction

Medication recommendation, as a vital technique, aims to assist doctors in prescribing effective medications while ensuring safety for patients' current visits, and has recently gained prominence in healthcare application.

Current medication recommendation methods [3, 13, 25, 32, 34, 37] typically analyze patients' medical histories using time series models like recurrent neural networks (RNNs). However, these models' limited hidden states hinder accurate recall of past facts. To address this, some studies [15, 23, 24, 26] use memory networks to encode and dynamically store historical records, while others [11, 14, 25, 33] utilize attention mechanism to capture relationships between records. Nevertheless, the normalization inherent in attention mechanism can distort these relationships, causing scaling losses. For instance, consider a patient with two historical visits. If the patient's current visit shows similarities of 0.1 and 0.2 with two previous visits, normalization may adjust these values to 0.475 and 0.525, inadvertently amplifying the importance of irrelevant visits. Therefore, it is crucial to eliminate the impact of irrelevant information, capturing the relationships between the sequence of patients' visits to assess their current health conditions.

Data extracted from the medical information mart for intensive care (MIMIC) [6, 7] reveal that most medication combinations in prescriptions still have potential side effects. Therefore, alleviating the latent risk of drug-drug interactions (DDI) is a challenging issue. Existing works, such as SafeDrug [31] and CausalMed [10], usually apply various strategies like piece-wise or simulated annealing methods to reduce DDI risks. However, their results are not satisfactory. As shown in Figure 1, we compare the performances of these two models, respectively, on Jaccard index and DDI rate [25]. The Jaccard index measures the similarity between the correct medication combination and the actual recommended one. The DDI rate refers to the rate of the number of conflicting medication pairs against the total number of recommended medication pairs. We observe that CausalMed achieves a higher Jaccard index than SafeDrug but also has a higher DDI rate. Additionally, DDI rate fluctuations are erratic during training process for both models,

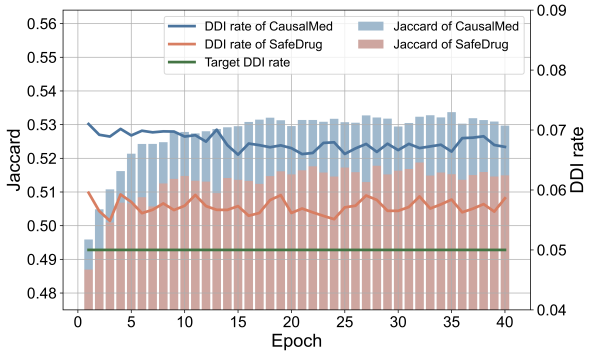


Figure 1: Comparison of performance between two models. A higher Jaccard index (denoting greater accuracy) and a lower DDI rate (indicating more safety) are preferable.

showing no discernible trend towards the target DDI rate (green line), as specified in the original studies. This uncontrollability emphasizes the necessity of developing more effective strategies to ensure the recommendation of appropriate medications with minimized DDI risks.

In this work, we aim to fill the research gap by developing a solution that is both accurate and safe. Towards this end, we build a new **G**raph augmented **P**atient **S**imilarity network for **Rec**ommendation, namely **GPSRec**. It separately processes patients' visits encoded by time series models and the raw collection of diagnosis and procedures, using dual similarity measures to selectively integrate historical visits with learned medication graphs and prescriptions, thereby effectively filtering irrelevant information to improve the accuracy of recommendation. Moreover, we present a novel training strategy to mitigate the risk of side effects from historical prescriptions. This strategy introduces a pre-training method to preliminary capture connections between medical entities and prioritize DDI risks. During the formal training, a dual threshold loss adjustment is implemented to further improve the safety.

The main contributions of this paper are as follows:

- We propose a new graph augmented patient similarity network that leverages dual similarity measures to effectively filter out irrelevant medical information, improving the accuracy of medication recommendation.
- We present a novel training strategy, which combines a pre-training method and a dual threshold loss adjustment, mitigating the potential risk of side effects in historical prescriptions, and effectively enhancing the safety of medication recommendation.
- We conduct comprehensive experiments utilizing two real datasets to evaluate the effectiveness of our proposed method. Extensive experiments demonstrate that our model significantly outperforms state-of-the-art models.

2 Related Work

Medication recommendation can be classified into two main categories: instance-based methods, which rely solely on the current visit, and longitudinal methods, which leverage historical visits for prediction. In instance-based methods, SMR [5] integrates electronic health records and medical knowledge via heterogeneous graphs for link prediction. RAREMed [35] adopts a pretrain-finetune for rare diseases. In longitudinal methods, KAMPNet [1] and CEHMR [19] capture implicit relationships from the medical ontology graph. DrugRec [17] and CasualMed [10] introduce causal models to find the relationships between different medical information. In addition, MICRON [30] predicts health changes using residual networks based solely on the prior visit. In contrast, some works [15, 24, 26, 33] utilize RNNs to model sequential visits. COGNet [27], Trans-GAHNet [11], DMRNet [16] and VITA [8] use a Transformer [22] to encode medical information, while NLA-MMR [20] leverages pre-trained language models for encoding. Among these, several methods [11, 15, 16, 20, 33] use attention mechanisms to retrieve relevant historical prescriptions for the current state, but indiscriminate attention risks amplifying irrelevant information and reducing accuracy. Additionally, most of the aforementioned methods primarily focus on the recommendation accuracy and fail to fully account for safety considerations. *In comparison, our work emphasizes the rational utilization of historical visits while reducing the DDI risk of recommended medication combinations, improving the accuracy and safety of recommendation.*

Graph attention networks (GATs) employ an attention mechanism [9] to compute relationships between nodes. Unlike traditional graph neural networks [28], which update each node's state independently, GATs consider the states of neighboring nodes during updates, capturing structural information more effectively. In social networks, GATs can identify community structures or predict user behavior [21, 29]. In medication recommendation, PREMIER [2] uses them to analyze co-occurrence and interaction information among medications. SafeDrug [31] and Camen [3] introduce a new method by encoding molecular graphs to differentiate medications. On this basis, D-GCAN [18] and ACDNet [14] employ GATs to calculate the importance of atoms and bonds. However, these methods rely on hyperparameters to integrate representations, potentially limiting differentiation within a single graph. *In this paper, our method reduces hyperparameter dependence, using an attention mechanism to adaptively fuse multi-graph medication features (e.g., molecular graphs, side effect relationship graphs, etc.).*

Avoiding adverse DDI in medication has recently attracted attention. Existing works typically leverage DDI information through various DDI loss adjustment strategies. For example, GAMENet [15] transforms different loss functions with calculated probabilities, but it underperforms. SafeDrug [31] employs a piece-wise strategy to maintain a lower DDI rate, but this significantly compromises accuracy. FFBDNet [25] uses a greedy strategy to mask high-conflict medications, which may inadvertently exclude potentially effective treatment options. MoleRec [32] utilizes a simulated annealing strategy to dynamically adjust parameters in an attempt to balance two competing losses, yet its optimization process fails to progress toward intended objectives. Additionally, RASNet [36] introduces an exponential controller to adjust the DDI loss, while FDIRNet [12] incorporates an interaction penalty. However, experimental results from these methods indicate that existing strategies still fall short in effectively mitigating the risk of side effects. *Unlike these, our method leverages pre-training and a dual-threshold loss strategy for precise DDI control, offering a new solution to enhance the safety of medication recommendation.*

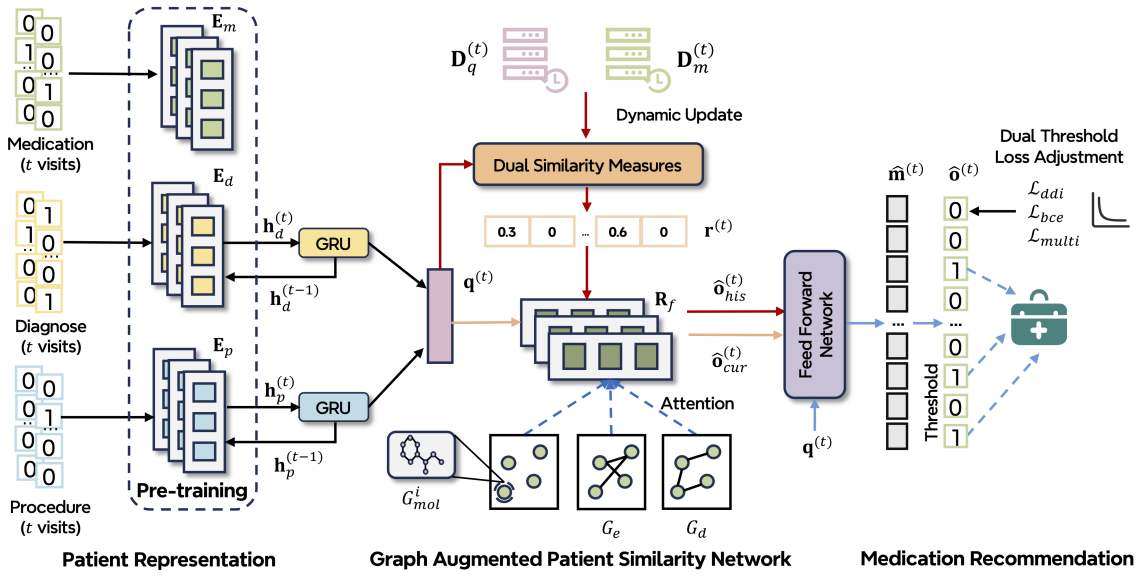


Figure 2: GPSRec: At the current t-th visit, we first utilize the pre-training module and patient representation module to encode a patient's complete visits and obtain a patient representation $\mathbf{q}^{(t)}$. Next, we apply GATs to model the EHR graph G_e, DDI graph G_d, and molecular graphs G^i_{mol}, generating a comprehensive medication representation $\mathbf{R}_f$ through an attention mechanism. The patient similarity network dynamically stores these information, performing a reasonable analysis of historical visits and mapping them to the medication graph representation to derive a historical output $\hat{\mathbf{o}}^{(t)}_{his}$. Simultaneously, the current visit information is processed to generate a current output $\hat{\mathbf{o}}^{(t)}_{cur}$. Finally, GPSRec combines the two outputs with the patient's current health condition $\mathbf{q}^{(t)}$ to recommend the most suitable medications.

3 Problem Formulation

Electronic Health Record (EHR). The preprocessing of EHR primarily involves longitudinal time series data represented by medical codes, which capture comprehensive records of each patient's medical history. Formally, each patient's EHR can be represented as a sequence $\mathbf{V} = [\mathbf{v}^{(1)}, \mathbf{v}^{(2)}, \ldots, \mathbf{v}^{(T)}]$, where T denotes the total number of visits for that patient. $\mathbf{v}^{(t)}$ records the information from the t-th visit, represented as $\mathbf{v}^{(t)} = [\mathbf{d}^{(t)}, \mathbf{p}^{(t)}, \mathbf{m}^{(t)}]$, where $\mathbf{d}^{(t)} \in \{0,1\}^{|\mathcal{D}|}$, $\mathbf{p}^{(t)} \in \{0,1\}^{|\mathcal{P}|}$ and $\mathbf{m}^{(t)} \in \{0,1\}^{|\mathcal{M}|}$ are multi-hot diagnosis, procedure and medication vectors, respectively. Here, $\mathcal{D}$, $\mathcal{P}$ and $\mathcal{M}$ represent the sets of medical codes, and $|\cdot|$ denotes the cardinality (the number of elements) of each set.

DDI Graph and EHR Graph. The DDI graph and EHR graph can be denoted as G_d and G_e respectively. We use a symmetric matrix $\mathbf{A}^d \in \{0,1\}^{|\mathcal{M}|\times|\mathcal{M}|}$ to represent the DDI graph G_d, where $\mathbf{A}^d_{ij} = 1$ indicates the existence of an adverse interaction between medications i and j. Similarly, we represent the EHR graph G_e using a symmetric matrix $\mathbf{A}^e \in \{0,1\}^{|\mathcal{M}|\times|\mathcal{M}|}$, where $\mathbf{A}^e_{ij} = 1$ signifies that medications i and j have appeared in the same prescription.

Molecular Graph. The molecular graph for medication i can be represented as G^i_{mol}. First, we collect a node set $C = \{a_1, a_2, \ldots, a_{|C|}\}$ of all appearing atoms in the graphs. Then, we design a learnable atom embedding table $\mathbf{E}_a \in \mathbb{R}^{|C|\times l}$, where l is the predefined size of the embedding and each row corresponds to the initial vector for a particular atom. Furthermore, we use $\mathbf{A}^{mol}_i$ to represent the existence of chemical bonds between the nodes in the molecular graph of medication i. The adjacency matrix dimensions of different molecular graphs may not necessarily be the same.

4 Proposed Model

As shown in Figure 2, GPSRec consists of four key modules: a pre-training module, a patient representation module, a graph augmented patient similarity network module, and a medication recommendation module.

4.1 Pre-training Module

To initially capture the complex entity associations between medications and visit information in the EHR, we employ a pre-training method that simulates the process of a doctor's visit to obtain initial representations.

Given two adjacent medical records of a patient, denoted as $\mathbf{v}^{(t-1)} = [\mathbf{d}^{(t-1)}, \mathbf{p}^{(t-1)}, \mathbf{m}^{(t-1)}]$ and $\mathbf{v}^{(t)} = [\mathbf{d}^{(t)}, \mathbf{p}^{(t)}, \mathbf{m}^{(t)}]$. Notably, if the patient has no historical visits, all three vectors in $\mathbf{v}^{(t-1)}$ are set to zero vectors. We employ three learnable embedding tables, $\mathbf{E}_d \in \mathbb{R}^{|\mathcal{D}|\times l}$, $\mathbf{E}_p \in \mathbb{R}^{|\mathcal{P}|\times l}$ and $\mathbf{E}_m \in \mathbb{R}^{|\mathcal{M}|\times l}$, corresponding to diagnoses, procedures and medications. Then, we map the $\mathbf{d}^{(t)}$, $\mathbf{p}^{(t)}$ and $\mathbf{m}^{(t-1)}$ to their corresponding embeddings to get the aggregate vectors, $\mathbf{e}^{(t)}_d$, $\mathbf{e}^{(t)}_p$ and $\mathbf{e}^{(t-1)}_m$, all of which belong to $\mathbb{R}^l$:

$$\mathbf{e}^{(t)}_d = \mathbf{d}^{(t)}\mathbf{E}_d, \mathbf{e}^{(t)}_p = \mathbf{p}^{(t)}\mathbf{E}_p, \mathbf{e}^{(t-1)}_m = \mathbf{m}^{(t-1)}\mathbf{E}_m. \quad (1)$$

We concatenate the three representations into a single vector to form the patient's preliminary representation, and finally pass

through a feed-forward network (FFN), resulting in the medication predicted probabilities $\hat{\mathbf{m}}_{pre}^{(t)} \in \mathbb{R}^{|\mathcal{M}|}$:

$$\hat{\mathbf{m}}_{pre}^{(t)} = \text{FF}_1(\text{CONCAT}(\mathbf{e}_d^{(t)}, \mathbf{e}_p^{(t)}, \mathbf{e}_m^{(t-1)})), \tag{2}$$

where $\text{FF}_1(\cdot) : \mathbb{R}^{3l} \rightarrow \mathbb{R}^{|\mathcal{M}|}$, consists of a fully connected layer followed by a sigmoid function.

During the pre-training phase, we prioritize reducing the DDI risks. Therefore, when adopting Eqs. (18) to (22) as our loss function, we use parameters different from formal training, as detailed in the experimental section.

Subsequently, we feed the obtained initial representations, $\mathbf{E}_d$, $\mathbf{E}_p$ and $\mathbf{E}_m$, which contain certain entity associations, into the subsequent module.

4.2 Patient Representation Module

In the pre-training phase, we utilize solely the previous prescriptions to represent prior visits. While this approach is straightforward, it provides limited information and may not fully assess the influence of previous visits on the current one. Therefore, during the formal training phase, we employ the time series models to further encode the medical information.

First, we leverage two learnable embedding tables $\mathbf{E}_d$ and $\mathbf{E}_p$, which are obtained from pre-training process, and continue to use Eq. (1) to match the diagnosis and procedure embeddings with the multi-hot vectors $\mathbf{d}^{(t)}$ and $\mathbf{p}^{(t)}$. Then, we apply gated recurrent neural networks (GRUs) to model both the diagnostic sequence and procedure sequence:

$$\mathbf{h}_d^{(t)} = \text{GRU}_d(\mathbf{e}_d^{(t)}, \mathbf{h}_d^{(t-1)}), \mathbf{h}_p^{(t)} = \text{GRU}_p(\mathbf{e}_p^{(t)}, \mathbf{h}_p^{(t-1)}), \tag{3}$$

where $\mathbf{h}_d^{(t)}, \mathbf{h}_p^{(t)} \in \mathbb{R}^l$ are hidden states of GRU_d and GRU_p, respectively. We initialize $\mathbf{h}_d^{(0)}$ and $\mathbf{h}_p^{(0)}$ as zero vectors, indicating no prior visit. Subsequently, we concatenate the updated hidden diagnosis state $\mathbf{h}_d^{(t)}$ and hidden procedure state $\mathbf{h}_p^{(t)}$ and pass through a FFN to derive a comprehensive patient representation $\mathbf{q}^{(t)} \in \mathbb{R}^l$:

$$\mathbf{q}^{(t)} = \text{FF}_2(\text{CONCAT}[\mathbf{h}_d^{(t)}, \mathbf{h}_p^{(t)}]), \tag{4}$$

where $\text{FF}_2(\cdot) : \mathbb{R}^{2l} \rightarrow \mathbb{R}^l$, is a fully connected layer.

4.3 Graph Augmented Patient Similarity Network Module

Medication Graph Representation Learning. GATs can effectively compute relationships between graph nodes, allowing for better capture of existing entity associations. The inputs to GATs include the initial embedding table of medication $\mathbf{E}_m$, obtained from the pre-training module, and the adjacency matrices $\mathbf{A}^e$ and $\mathbf{A}^d$. These are utilized within GATs to update the feature representation of each medication:

$$\mathbf{R}_e = \text{GAT}(\mathbf{E}_m, \mathbf{A}^e), \mathbf{R}_d = \text{GAT}(\mathbf{E}_m, \mathbf{A}^d). \tag{5}$$

After that, we obtain the graph representation $\mathbf{R}_e \in \mathbb{R}^{|\mathcal{M}| \times l}$ for the EHR graph and the graph representation $\mathbf{R}_d \in \mathbb{R}^{|\mathcal{M}| \times l}$ for the DDI graph.

For the graph representation learning $\mathbf{R}_{mol}$ of the molecular graphs, our operation is different from the previous graphs. We need to learn and summarize the molecular graph of each medication separately to obtain a comprehensive representation that encompasses all medications.

Given a molecular graph G_{mol}^i, we extract the set of atoms $C_i' = \{a_1, \ldots, a_n\}$ from this graph, where $n \leq |C|$, and an adjacency matrix $\mathbf{A}_i^{mol} \in \{0,1\}^{|C_i'| \times |C_i'|}$ is used to represent the existence of chemical bonds between the atoms in G_{mol}^i. Then we use GATs and *readout* function [4], denoted as $\text{READOUT}(\cdot)$, to compute the representation of a medication molecule $\mathbf{R}_i^{mol} \in \mathbb{R}^l$:

$$\mathbf{R}_i^{mol} = \text{READOUT}(\text{GAT}(\mathbf{E}_a, \mathbf{A}_i^{mol})). \tag{6}$$

By learning the molecular graph of each medication, we finally obtain a complete molecular representation $\mathbf{R}_{mol} \in \mathbb{R}^{|\mathcal{M}| \times l}$:

$$\mathbf{R}_{mol} = [\mathbf{R}_1^{mol} || \mathbf{R}_2^{mol} || \ldots || \mathbf{R}_{|\mathcal{M}|}^{mol}]. \tag{7}$$

We think that sharing the same parameters across all medications during fusion could lead to equal amplification or reduction of information in a single graph. Instead, we use an attention mechanism to analyze information from distinct graphs for each medication.

We compute the product between the query vector $\mathbf{q}^{(t)} \in \mathbb{R}^l$ and each of the key embedding tables $\mathbf{R}_*$ (where $\mathbf{R}_*$ represents the graph representations $\mathbf{R}_e$, $\mathbf{R}_d$ and $\mathbf{R}_{mol}$), and then calculate the normalized weight of each medication in these graph representations:

$$\alpha_{*i} = \frac{exp(\mathbf{q}^{(t)} \mathbf{R}_{*i}^{\mathsf{T}})}{\sum_{c \in \{e,d,mol\}} exp(\mathbf{q}^{(t)} \mathbf{R}_{ci}^{\mathsf{T}})}, \tag{8}$$

where i ranges from 1 to $|\mathcal{M}|$, α_{*i} represents the normalized weight of the i-th medication in the graph representation $\mathbf{R}_*$. After calculation, we get three normalized weight vectors α_e, α_d and α_{mol}, all of which belong to $\mathbb{R}^{|\mathcal{M}|}$.

To ensure the correct dimensions for the subsequent matrix operations, we convert α_* into diagonal matrices with the same dimension as $\mathbb{R}^{|\mathcal{M}| \times |\mathcal{M}|}$ using the $diag(\cdot)$ function. Subsequently, we get the fused medication representation $\mathbf{R}_f \in \mathbb{R}^{|\mathcal{M}| \times l}$:

$$\mathbf{R}_f = \sum_{* \in \{e,d,mol\}} diag(\alpha_*) \mathbf{R}_* + \lambda \mathbf{E}_m, \tag{9}$$

where λ is a learnable hyperparameter. Adding the initial representation enriches the feature information.

Dual Similarity Measures. To compensate for the shortcomings of GRUs in information storage, we introduce a dynamic memory component $\mathbf{D}^{(t)}$ to store the patient's historical visit representations and prescriptions (as multi-hot vectors) in the form of key-value pairs. We incrementally insert key-value pair after each visit step. The component is represented as follows:

$$\mathbf{D}^{(t)} = \{\mathbf{q}^{(x)} : \mathbf{m}^{(x)}\}_{x=1}^{x=t-1}. \tag{10}$$

For clarity, we use $\mathbf{D}_q^{(t)} = [\mathbf{q}^{(1)} || \ldots || \mathbf{q}^{(t-1)}] \in \mathbb{R}^{|t-1| \times l}$ to denote the key vectors and $\mathbf{D}_m^{(t)} = [\mathbf{m}^{(1)} || \ldots || \mathbf{m}^{(t-1)}] \in \mathbb{R}^{|t-1| \times |\mathcal{M}|}$ to denote the value vectors.

When capturing relationships between historical visits, using an attention mechanism for normalization can amplify similarities even when the current visit is not similar to historical visits. This is inappropriate because the similarity associated with unrelated historical visits should not be exaggerated. To address this, we no longer use the attention mechanism as employed in FFBDNet [25]

and DMRNet [16], but instead utilize dual similarity measures to capture relationships. Here, we clarify that the values of x in Eqs. (11) to (13) range from 1 to $t-1$.

First, we compute the similarity using the query vector (a fusion of diagnosis and procedure) $\mathbf{q}^{(x)} \in \mathbb{R}^l$ in $\mathbf{D}_q^{(t)}$:

$$s_{x,t}^{(q)} = \frac{\mathbf{q}^{(x)} \cdot \mathbf{q}^{(t)}}{|\mathbf{q}^{(x)}| \cdot |\mathbf{q}^{(t)}|}. \tag{11}$$

However, due to potential information loss in the fused representation $\mathbf{q}^{(x)}$ obtained from GRUs, we use the original diagnostic and procedure sets to compensate for this loss:

$$s_{x,t}^{(d)} = \frac{|\{\mathbf{d}^{(x)}\} \cap \{\mathbf{d}^{(t)}\}|}{|\{\mathbf{d}^{(x)}\} \cup \{\mathbf{d}^{(t)}\}|}, s_{x,t}^{(p)} = \frac{|\{\mathbf{p}^{(x)}\} \cap \{\mathbf{p}^{(t)}\}|}{|\{\mathbf{p}^{(x)}\} \cup \{\mathbf{p}^{(t)}\}|}, \tag{12}$$

where $\{\cdot\}$ represents a set of medical codes, which consists of the indices of elements with a value of 1 in a vector. We then combine the similarities of the two groups, considering the influence of visit order when setting weights:

$$r_x^{(t)} = \sum_{*\in\{q,d,p\}} exp\left(\lambda_* \frac{t-x}{t}\right) s_{x,t}^{(*)}, \tag{13}$$

where λ_* are learnable hyperparameters. $r_x^{(t)}$ is the x-th element in $\mathbf{r}^{(t)} \in \mathbb{R}^{|t-1|}$, which represents the similarity between the current visit and the x-th historical visit. We then map the computed similarities onto the medication graph representation, enabling accurate extraction of the historical output $\hat{\mathbf{o}}_{his}^{(t)} \in \mathbb{R}^l$:

$$\hat{\mathbf{o}}_{his}^{(t)} = \mathbf{r}^{(t)} \mathbf{D}_m^{(t)} \mathbf{R}_f. \tag{14}$$

4.4 Medication Recommendation Module

For the medication recommendation, we consider both current and historical visits. Notably, if a patient has no historical visits, $\hat{\mathbf{o}}_{his}^{(t)}$ is set to be the same as the current output $\hat{\mathbf{o}}_{cur}^{(t)} \in \mathbb{R}^l$, which is obtained as follows:

$$\hat{\mathbf{o}}_{cur}^{(t)} = \mathrm{FF}_3(\mathbf{q}^{(t)})\mathbf{R}_f, \tag{15}$$

where $\mathrm{FF}_3(\cdot) : \mathbb{R}^l \to \mathbb{R}^{|\mathcal{M}|}$, is a fully connected layer. By combining the historical output $\hat{\mathbf{o}}_{his}^{(t)}$, we obtain the recommended probabilities of medications, denoted as $\hat{\mathbf{m}}^{(t)} \in \mathbb{R}^{|\mathcal{M}|}$, as follows:

$$\hat{\mathbf{m}}^{(t)} = \mathrm{FF}_4(\mathrm{CONCAT}[\mathbf{q}^{(t)}, \hat{\mathbf{o}}_{cur}^{(t)}, \hat{\mathbf{o}}_{his}^{(t)}]), \tag{16}$$

where $\mathrm{FF}_4(\cdot) : \mathbb{R}^{3l} \to \mathbb{R}^{|\mathcal{M}|}$, consists of a fully connected layer followed by a sigmoid function. Subsequently, we obtain a prediction vector $\hat{\mathbf{o}}^{(t)} \in \{0,1\}^{|\mathcal{M}|}$ by selecting the items of $\hat{\mathbf{m}}^{(t)}$ whose value is greater than a predefined threshold value δ. The value of i ranges from 1 to $|\mathcal{M}|$.

$$\hat{\mathbf{o}}^{(t)} = \begin{cases} 1 & \hat{\mathbf{m}}_i^{(t)} \geq \delta \\ 0 & \hat{\mathbf{m}}_i^{(t)} < \delta \end{cases}. \tag{17}$$

4.5 Training and Inference

In this section, we propose a novel adjustment strategy that enables the model to find better accuracy under conditions of a lower DDI rate.

Multiple Loss Function. Our method for the recommendation task involves treating it as a multi-label classification task. To achieve this, we utilize two commonly used loss functions: the binary cross-entropy loss, denoted as $\mathcal{L}_{bce}$, and the multi-label margin loss, denoted as $\mathcal{L}_{multi}$:

$$\mathcal{L}_{bce} = -\sum_{i=1}^{|\mathcal{M}|} \mathbf{o}_i^{(t)} \log(\hat{\mathbf{m}}_i^{(t)}) + (1 - \mathbf{o}_i^{(t)}) \log(1 - \hat{\mathbf{m}}_i^{(t)}), \tag{18}$$

$$\mathcal{L}_{multi} = \sum_{\mathbf{o}_i^{(t)}=1, \mathbf{o}_j^{(t)}=0} \frac{\max(0, 1 - (\hat{\mathbf{m}}_i^{(t)} - \hat{\mathbf{m}}_j^{(t)}))}{|\mathcal{M}|}, \tag{19}$$

where $\mathbf{o}^t \in \mathbb{R}^{|\mathcal{M}|}$ represents the ground truth of the medication combination.

DDI Loss Function. We define the DDI loss function as follows:

$$\mathcal{L}_{ddi} = \sum_t \sum_{i=1}^{|\mathcal{M}|} \sum_{j=1}^{|\mathcal{M}|} (\hat{\mathbf{m}}_i^{(t)} \cdot \hat{\mathbf{m}}_j^{(t)}) \cdot \mathbf{A}_{ij}^d, \tag{20}$$

where $\mathbf{A}_{ij}^d$ represents the medication adverse DDI adjacency matrix.

Combined Controllable Loss Function. We adopt the weighted sum form of combined loss:

$$\mathcal{L} = u \cdot (v \cdot \mathcal{L}_{bce} + (1-v) \cdot \mathcal{L}_{multi}) + (1-u) \cdot \mathcal{L}_{ddi}, \tag{21}$$

where u and v are hyperparameters. Clinical datasets reveal that most medication combinations carry potential side effects [6, 7]. Blindly minimizing prediction error may propagate these risks, while over-constraining DDI could reduce accuracy. Therefore, we address this dilemma through phased optimization.

Dual Threshold Loss Adjustment. Building upon the pre-training phase described in Section 4.1, our model captures fundamental medication-disease relationships. During the formal training process, we present a dual threshold loss adjustment strategy to control the variation of hyperparameter u. This strategy accommodates the effects of pre-training and further optimizes model performance.

$$u = \begin{cases} 1 & \rho \leq \theta_2 \\ \exp\left(-\left(\frac{|\rho-\theta_1|}{\gamma} + \epsilon\right)\right) & \rho > \theta_2 \end{cases}, \tag{22}$$

where ρ represents the actual DDI rate, we first set $\theta_1 = 0.05$, aligning with prior works [10, 31, 32], as our target DDI rate. Due to the pre-training method, both the initial DDI rate and accuracy are relatively low, as evidenced by the experimental results in Section 5.5, providing more room to improve accuracy. However, while enhancing accuracy, the DDI risks must still be considered. Therefore, we set the threshold θ_2, which determines when to incorporate $\mathcal{L}_{ddi}$ into consideration, at a level lower than θ_1. This configuration is necessary since the initial DDI rate already falls below θ_1.

Additionally, we introduce the expression $\exp\left(-\left(\frac{|\rho-\theta_1|}{\gamma} + \epsilon\right)\right)$ to stabilize and facilitate the training process. Unlike the strategy proposed in [10, 31, 32], our method leverages the lower DDI risks obtained through pre-training as a foundation. To ensure that the weight adjustment remains within a controlled range, we introduce the hyperparameter γ, guiding the actual DDI rate ρ to gradually align with the target rate θ_1. The correction term ϵ is incorporated to smoothly modulate the adjustment of u. The complete training procedure is detailed in algorithm 1.

Algorithm 1 Training Process for GPSRec

Input: Training set $\mathbf{V} = [\mathbf{v}^{(1)}, \mathbf{v}^{(2)}, \ldots, \mathbf{v}^{(T)}]$, the adjacency matrices $\mathbf{A}^e, \mathbf{A}^d$ and $\mathbf{A}_i^{mol}$ and training epoches L.
Output: The recommendation results for patient $\hat{\mathbf{o}}^{(t)}$.

Generate the initial representations $\mathbf{E}_d, \mathbf{E}_p$ and $\mathbf{E}_m$ according to Eqs. (1)-(2) by a pre-training method.
for $k = 1$ to L **do**
  for patient $j = 1$ to N **do**
    $\mathcal{L} \leftarrow 0$
    for historical visit $t = 1$ to T **do**
      Generate the query vector $\mathbf{q}^{(t)}$ by Eqs. (3)-(4).
      Generate the representations for the graphs $\mathbf{R}_e, \mathbf{R}_d$ and $\mathbf{R}_{mol}$ by Eqs. (5)-(7).
      Generate the representation $\mathbf{R}_f$ by Eqs. (8)-(9).
      Calculate the similarity sequence $\mathbf{r}^{(t)}$ through a dual similarity measures by Eqs. (10)-(13).
      Generate the historical output $\hat{\mathbf{o}}_{his}^{(t)}$ and the current output $\hat{\mathbf{o}}_{cur}^{(t)}$ by Eqs. (14)-(15).
      Generate the predicted probability $\hat{\mathbf{m}}^{(t)}$ and the predicted output $\hat{\mathbf{o}}^{(t)}$ by Eqs. (16)-(17).
      Calculate $\mathcal{L}_{bce}, \mathcal{L}_{multi}$ and $\mathcal{L}_{ddi}$ by Eqs. (18)-(20).
      Accumulate the loss $\mathcal{L}$ by Eqs. (21)-(22).
    end for
    Update the learnable parameters.
  end for
end for
return $\hat{\mathbf{o}}^{(t)}$.

Efficiency. To clarify the computational efficiency of GPSRec, we analyze the time complexity of each module and simplify the expressions to their asymptotically dominant terms.

The **pre-training module** processes T historical visits per patient, where each visit includes an embedding lookup and a forward pass over diagnoses ($|\mathcal{D}|$), procedures ($|\mathcal{P}|$) and medications ($|\mathcal{M}|$). In the worst case, this costs $O\Big(T \times \big((|\mathcal{D}|+|\mathcal{P}|+|\mathcal{M}|) \times l + |\mathcal{M}| \times l\big)\Big)$. According to the clinical statistics in Table 1, where $|\mathcal{D}| > |\mathcal{P}|$ and $|\mathcal{D}| > |\mathcal{M}|$, the time complexity simplifies to $O\big(T \times |\mathcal{D}| \times l\big)$. The **patient representation module** employs two GRUs over T time steps, resulting in $O\big(T \times l^2\big)$ for hidden-state updates. In the **graph augmented patient similarity network module**, processing the EHR/DDI graph via GATs requires $O\left(|\mathcal{M}|^2 \times l\right)$, driven by pairwise medication interactions. For molecular graph operations-assuming a bounded maximum number of atoms (C) per medication-the complexity reduces to $O\left(|\mathcal{M}| \times l\right)$, with C^2 absorbed as a constant. Computing **the dual similarity measures** and **generating recommendations** for each visit requires $O\big(T \times |\mathcal{M}| + |\mathcal{M}| \times l\big)$.

By dropping all strictly lower-order terms, the overall time complexity reduces to $O\Big(|\mathcal{M}|^2 \times l + T \times (|\mathcal{D}| \times l + l^2 + |\mathcal{M}|)\Big)$. This formulation highlights two key scalability factors: a quadratic dependence on the number of medications for graph interactions, and linear scaling with visits and medications for patient-specific computations-ensuring GPSRec's efficiency on large-scale clinical data.

5 Experiments

In this section, we conduct comprehensive experiments to answer the following five questions:

- **RQ1**: How does the performance of the proposed GPSRec compare to that of existing medication recommendation models?
- **RQ2**: How do the individual components of the GPSRec influence its performance?
- **RQ3**: Does the GPSRec effectively filter out irrelevant medical information?
- **RQ4**: Does the GPSRec improve the safety of medication recommendation?
- **RQ5**: Can GPSRec be effective in practical clinical scenarios?

5.1 Experimental Setup

Datasets. We evaluate our model on MIMIC-III [6] and MIMIC-IV [7]. We split train-validation-test by $\frac{2}{3} - \frac{1}{6} - \frac{1}{6}$. After preprocessing, the statistics of two datasets can be shown in Table 1.

Table 1: The statistics of the datasets.

Items	MIMIC-III	MIMIC-IV
# of visits	15,032	166,872
# of patients	6,350	64,273
# of diagnosis	1,958	1,994
# of procedure	1,430	1,499
# of medication	131	131
max/avg.# of diagnoses	128/10.51	267/8.24
max/avg.# of procedures	50/3.84	64/2.18
max/avg.# of medications	65/11.65	72/6.67
max/avg.# of visits	29/2.37	70/2.59

Experimental Environment. Experiments are conducted on an Ubuntu 20.04 workstation with 12 CPUs, 40GB memory and an NVIDIA RTX 3080 GPU (10GB VRAM). PyTorch 2.0.0 and CUDA 11.8 are used to support the research.
Evaluation Metrics. We measure the performance with four common metrics, including DDI rate, Jaccard index, Precision-Recall AUC (PRAUC) and F1-score. The calculation equations for evaluation indicators can refer to [32].
Baselines. To evaluate our work comprehensively, we compare GPSRec with the state-of-the-art methods, including GAMENet [15], SafeDrug [31], COGNet [27], FFBDNet [25], MoleRec [32], VITA [8], RASNet [36], DMRNet [16] and CausalMed [10].
Implementation Details. Throughout the training phase, we set the embedding dimension l to 64 and the threshold δ to 0.5. During pre-training, the loss function parameters are set to $v = 0.95$, $\theta_1 = 0.05$, $\theta_2 = 0.03$ and $\gamma = 0.67$, with the preset correction term ϵ set to 2. In the patient representation module, both GRU_d and GRU_p are configured with 64 hidden units. In the graph augmented patient similarity network module, the GATs have a dropout rate of 0.2. During the formal training phase, we set $v = 0.95$, $\theta_1 = 0.05$, $\theta_2 = 0.01$ and $\gamma = 0.50$, with the preset correction term ϵ set to 2.04. The pre-training and formal training last for 15 and 25 epochs, respectively, and the Adam optimizer is employed with a learning rate of $lr = 0.0005$.

Table 2: Performance comparison on MIMIC-III and MIMIC-IV. Bold numbers indicate the best performance.

Model	MIMIC-III				MIMIC-IV			
	DDI Rate	Jaccard	PRAUC	F1-score	DDI Rate	Jaccard	PRAUC	F1-score
GAMENet	0.0842	0.5204	0.7717	0.6755	0.0905	0.4502	0.6810	0.6041
SafeDrug	0.0591	0.5173	0.7657	0.6736	0.0574	0.4498	0.6961	0.6025
COGNet	0.0858	0.5316	0.7707	0.6644	0.0914	0.4744	0.6814	0.6223
FFBDNet	0.0788	0.5307	0.7764	0.6820	0.0740	0.4701	0.7094	0.6203
MoleRec	0.0726	0.5310	0.7758	0.6847	0.0731	0.4724	0.7119	0.6253
VITA	0.0781	0.5301	0.7775	0.6841	0.0710	0.4715	0.7096	0.6211
RASNet	0.0602	0.5321	0.7801	0.6882	0.0587	0.4731	0.7125	0.6249
DMRNet	0.0670	0.5319	0.7762	0.6859	0.0684	0.4740	0.7137	0.6261
CausalMed	0.0694	0.5332	0.7798	0.6860	0.0662	0.4754	0.7140	0.6255
GPSRec (Ours)	**0.0485**	**0.5390**	**0.7874**	**0.6913**	**0.0497**	**0.4807**	**0.7190**	**0.6315**

Table 3: Model Efficiency Comparison.

Model	Training Time /Epoch(s)	Best Epoch	Total Training Time(s)
GAMENet	47.52	42	1995.84
SafeDrug	42.56	47	2000.32
COGNet	40.60	107	4344.20
FFBDNet	162.76	17	2766.92
MoleRec	273.07	19	5188.33
VITA	134.09	52	7106.77
RASNet	42.24	83	3505.92
DMRNet	-	-	-
CausalMed	173.21	35	6062.35
Pre-training	31.40	7	219.80
GPSRec w/o *pre*	152.10	20	3042.00
GPSRec	-	-	3261.80

5.2 Overall Performance Comparison (RQ1)

GPSRec achieves superior accuracy compared to existing methods. To evaluate GPSRec's clinical performance, accuracy metrics (Jaccard, PRAUC and F1-score) and safety indicators (DDI rate) are employed, with all models tested five times using different seeds for robustness. As shown in Table 2, GPSRec consistently outperforms existing methods across all accuracy metrics. On MIMIC-III, it achieves improvements of 1.08% (Jaccard), 0.97% (PRAUC) and 0.77% (F1-score) over the state-of-the-art model, CausalMed. Similarly, on MIMIC-IV, it achieves gains of 1.11%, 0.70% and 0.95%, respectively. These improvements arise from two innovations: 1) selective integration of relevant historical visits through dual similarity measures, which filters out irrelevant records and enhances alignment with the patient's current health state, and 2) a two-stage training method that decouples accuracy optimization from safety constraints, achieving refined performance in both aspects. By selectively utilizing contextually relevant historical data, GPSRec maximizes its predictive accuracy while ensuring safety.

GPSRec achieves significantly enhanced safety compared to previous models. As shown in Table 2, GPSRec reduces DDI rates by 30.11% on MIMIC-III and 24.92% on MIMIC-IV, outperforming state-of-the-art models like CausalMed. History-reliant models (e.g., GAMENet, COGNet) increase prescription risks by indiscriminately using historical data, while molecular-focused methods (e.g., SafeDrug, MoleRec) prioritize safety at the cost of accuracy. VITA and DMRNet face error accumulation due to their cascaded architectures. Although RASNet applies a new loss function to reduce DDI risks, its DDI rates (0.0602/0.0587) remain higher than GPSRec's (0.0485/0.0497). In contrast, GPSRec incorporates a safety-centric pre-training phase to initialize low-risk embeddings and uses a dual threshold strategy to optimize accuracy and safety simultaneously.

GPSRec achieves greater computational efficiency, completing training in significantly less time. As shown in Table 3, GPSRec completes training in 3,261.80 seconds, which is 46.2% faster than CausalMed and 54.1% faster than VITA. This efficiency stems from two key innovations: 1) similarity-based filtering with linear complexity to eliminate the quadratic overhead of attention mechanisms, and 2) pre-training to achieve stable embedding initialization, which reduces training iterations. These strategies not only enhance computational speed but also improve training stability, ensuring consistent performance across varied datasets. Additionally, GPSRec is tested on commonly available hardware, including an NVIDIA RTX 3080 with 10GB VRAM, demonstrating its practicality for real-world clinical deployment without requiring specialized equipment or excessive resources.

5.3 Ablation Study (RQ2)

To evaluate the effectiveness of each component in GPSRec, we conduct an ablation study with the following ablation models:

- GPSRec w/o *dual*: we replace the dual similarity measures with an attention mechanism.
- GPSRec w/o $\hat{\mathbf{o}}_{his}^{(t)}$: We remove the graph augmented patient similarity network module and only use the current visit information.
- GPSRec w/o $\mathbf{R}_f$: We remove the attention mechanism, and adopt a linear summation for representation fusion.
- GPSRec w/o *pre*: We remove the pre-training phase and randomly initialize the original embedding tables.
- GPSRec w/o *strategy*: We remove the dynamic loss function and utilize the same loss function as CausalMed [10].

Table 4: Ablation study on MIMIC-III and MIMIC-IV. Bold numbers indicate the best performance.

Model	MIMIC-III				MIMIC-IV			
	DDI Rate	Jaccard	PRAUC	F1-score	DDI Rate	Jaccard	PRAUC	F1-score
GPSRec *w/o dual*	0.0513	0.5330	0.7820	0.6867	0.0515	0.4782	0.7132	0.6297
GPSRec *w/o* $\hat{\mathbf{o}}_{his}^{(t)}$	0.0492	0.5309	0.7829	0.6847	0.0500	0.4751	0.7118	0.6258
GPSRec *w/o* $\mathbf{R}_f$	0.0541	0.5312	0.7830	0.6851	0.0530	0.4768	0.7125	0.6271
GPSRec *w/o pre*	0.0517	0.5224	0.7735	0.6780	0.0508	0.4679	0.7073	0.6201
GPSRec *w/o strategy*	0.0669	0.5326	0.7818	0.6861	0.0657	0.4759	0.7114	0.6283
GPSRec (Ours)	**0.0485**	**0.5390**	**0.7874**	**0.6913**	**0.0497**	**0.4807**	**0.7190**	**0.6315**

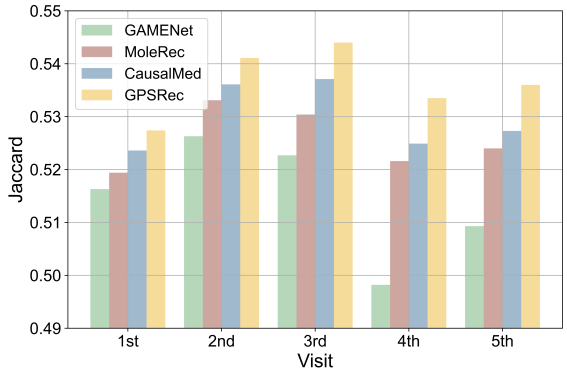


Figure 3: The performance of different models across multiple visits on MIMIC-III.

As shown in Table 4, GPSRec *w/o dual* demonstrates that using the attention mechanism to calculate relationships between medical visits negatively impacts accuracy, confirming our earlier hypothesis about attention's tendency to amplify irrelevant information. In contrast, GPSRec *w/o* $\hat{\mathbf{o}}_{his}^{(t)}$ verifies the value of our proposed network, which enables the model to more effectively incorporate the influence of past visits for accurate medication recommendation. Additionally, GPSRec *w/o* $\mathbf{R}_f$ demonstrates the benefits of medication-specific learning weights. This approach allows the model to better differentiate the unique characteristics (e.g., molecular features and side effects) of each medication, highlighting the effectiveness of using graphs to augment medication recommendation. Furthermore, GPSRec *w/o pre* illustrates the importance of the pre-training phase, which is essential for enhancing model stability and establishing a robust foundation for subsequent training. The results for GPSRec *w/o strategy* validate the efficacy of our novel training strategy. Our pre-training phase ensures a low initial DDI rate. This approach allows the model to identify safer medication combinations early, addressing a gap left by existing works such as SafeDrug or CausalMed. Notably, even without pre-training, GPSRec can initialize embeddings randomly and learn domain-specific representations through end-to-end training. The results show that GPSRec achieves competitive performance (DDI Rate: 0.0517/0.0508 on MIMIC-III/MIMIC-IV), demonstrating its flexibility in domain adaptation.

Table 5: Comparison of filtering methods.

Patient Case	Historical Visit	$\mathcal{D}/\mathcal{P}$ Jaccard	$\mathcal{M}$ Jaccard	GPSRec *w dual*	GPSRec *w att*
1	Visit 1	0.82	0.48	0.62	1.00
2	Visit 1	0.60	0.42	0.38	0.57
	Visit 2	0.78	0.55	0.75	0.43
3	Visit 1	0.43	0.37	0.25	0.27
	Visit 2	0.62	0.50	0.57	0.36
	Visit 3	0.85	0.58	0.72	0.37

5.4 Information Filtering Evaluation (RQ3)

To validate the superiority of GPSRec's dual similarity mechanism over conventional attention approaches, we conduct comparative analysis through weight allocation patterns and multi-visit performance. In Table 5, each row represents the relevance assessment between a historical visit and the current visit, where $\mathcal{D}/\mathcal{P}$ and $\mathcal{M}$ Jaccard represent the Jaccard similarity for diagnoses/procedures and medications respectively. GPSRec *w dual* uses dual similarity measures, and GPSRec *w att* employs an attention mechanism.

As evidenced in Table 5, the Softmax-constrained attention mechanism produces counterintuitive weight distributions. For single-visit cases, attention weights are forcibly normalized to 1.00 regardless of actual relevance, while our method provides dynamic assessment (weight=0.62) based on a combination of Jaccard index and semantic evaluation. In 2-visit cases, attention assigns questionable weights, whereas dual similarity suppresses noise through joint evaluation (Visit 2 weight=0.75 vs. attention's 0.43). This advantage becomes particularly evident in multi-visit scenarios: for high correlation records such as Visit 3 in 3-visit cases, dual similarity is given higher weight (weight=0.72 vs. attention's 0.37). Our method effectively avoids attention's normalization bias in medical record processing, particularly significant given that attention mechanisms intrinsically overweight single historical visits (fixed weight=1.00) regardless of actual clinical relevance.

Additionally, the advantages are further highlighted when comparing performance under varying visit frequencies. As Figure 3 demonstrates, GPSRec achieves superior prediction accuracy consistently across all visit frequencies, addressing the critical limitation of attention-based models, which often overemphasize irrelevant medications from initial visits. The consistent results observed in

the weight analysis, visit-frequency evaluation, and component ablation collectively validate the effectiveness of our method.

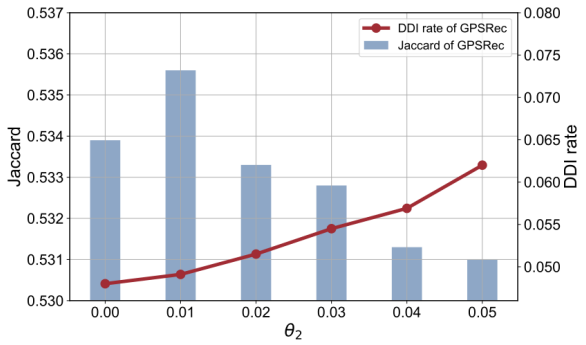


Figure 4: Impact of θ_2 values on MIMIC-III.

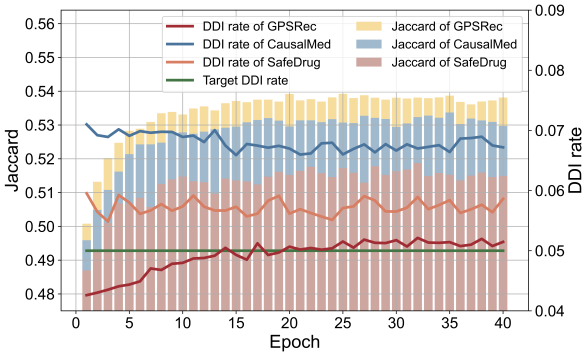


Figure 5: Comparison of performance between our model and the other two models on MIMIC-III.

5.5 Safety Performance Analysis (RQ4)

We conduct a sensitivity analysis on the values of θ_2 to evaluate the effectiveness of our proposed dual threshold loss adjustment strategy. To ensure a fair comparison, we set $\theta_1 = 0.05$, aligning it with the target DDI rate used in prior works [10, 31]. Figure 4 illustrates the impact of different values of θ_2 for model performance on MIMIC-III, validating the efficacy of our strategy. Specifically, GPSRec exhibits significant variations in DDI rates under different thresholds, demonstrating that the threshold selection allows the model to improve accuracy while maintaining DDI risk within an acceptable range. Meanwhile, we analyze the influence of the hyperparameter γ during the formal training phase. The results show that lower γ values (e.g., 0.3-0.5) accelerate convergence to the target DDI rate, while higher γ values (e.g., >0.6) provide more stable accuracy. This indicates that GPSRec is robust to reasonable variations in hyperparameter settings.

Additionally, Figure 5 shows a comparison of GPSRec with SafeDrug, the current state-of-the-art safety model, and another baseline, CausalMed. Notably, both baseline models use their own strategies to control the DDI risk. The results show that GPSRec achieves

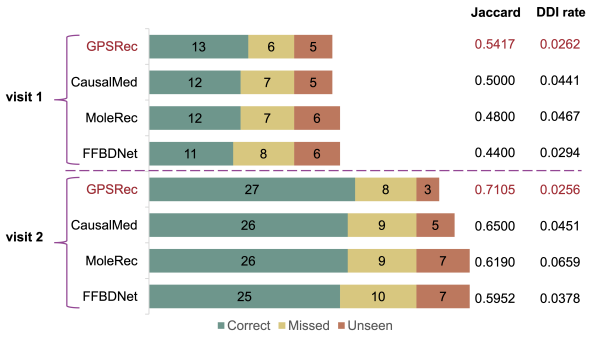


Figure 6: Recommended medications of a patient.

the lowest DDI rate while maintaining the highest accuracy compared to the other two baselines. Furthermore, by observing the trend of DDI rates during training, GPSRec's DDI rate gradually converges and stabilizes near the predefined target value as training progresses. This further confirms the effectiveness of our method in DDI risk control. Overall, these findings demonstrate that GPSRec can effectively enhance the safety of medication recommendation.

5.6 Case Study (RQ5)

We present a case study to showcase the effectiveness of our method. A patient from the test set with two visits is selected, and we compare the medication combinations predicted by GPSRec with those predicted by FFBDNet, MoleRec and CausalMed. In the bar charts, "correct" indicates medications that match the ground truth, "missed" denotes medications that are part of the ground truth but are not predicted by the models, and "unseen" represents medications predicted by the models but not present in the ground truth. As clearly shown in Figure 6, GPSRec predicts the highest number of correct medications while consistently minimizing incorrect predictions, as evidenced by the lowest "unseen" results. Notably, GPSRec achieves the highest Jaccard index across the two visits and recommends medication combinations with significantly lower DDI rates compared to the other models. This case study highlights the superiority of our approach, demonstrating that prioritizing both accuracy and safety in medication recommendation yields more optimal results than focusing solely on accuracy. Furthermore, it proves that GPSRec can recommend more accurate and safer medication combinations in practical clinical scenarios.

6 Conclusion

In this paper, we propose a novel model called GPSRec for medication recommendation. GPSRec leverages a graph augmented patient similarity network that selectively uses historical visits, effectively filtering irrelevant medical information to improve recommendation accuracy. Moreover, our new training strategy, which combines a pre-training method and a dual threshold loss adjustment, effectively mitigates the risk of adverse side effects and enhances recommendation safety. Experiments on two real-world datasets demonstrate that GPSRec outperforms state-of-the-art models significantly. In future work, we plan to extend GPSRec by incorporating additional contextual data (e.g., demographic data) to bridge the gap between computational predictions and safe medical practice.

GenAI Usage Disclosure

In this study, generative AI tools were solely employed to correct grammatical errors in the manuscript. No other uses of these tools occurred during the research process or content generation.

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